API Documentation

1. Authentication & Authorization APIs

Base URL: **http://localhost:5000/api/**

Description: Handles secure access using JWT and Bcrypt hashing.

POST /register

Description: Creates a new user account.

Authentication: Public.

Request Headers: Content-Type: application/json

Request Body Parameters:

- username (String): Unique identifier.
- email (String): Valid email address.
- password (String): Plain text password.
- role (String): **One of (**community_member, facility_manager, worker, admin.)

Example Request:

{ "username": "admin1", "email": "admin1@example.com", "password": "123456", "role":"admin"}

Example Success Response (201 Created):

{ "message": "User registered successfully", "user": { "user_id": 6, "username": "admin1", "email": "admin1@example.com" , "role": "admin", "is_active": true,"created_at": "2026-05-10T12:19:47.873Z" }}

POST /login

Description: Authenticates user and returns a JWT token.

Authentication: Public.

Request Body Parameters:

- • username (String)
- • password (String)

Example Request:

{ "email": "admin1@example.com", "password": "123456"}}

Example Success Response (200 OK):

{ "message": "Login successful", "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6Niwicm9sZSI6ImFkbWluIiwiaWF0IjoxNzc4NDI2Mzk1LCJleHAiOjE3Nzg0MzM1OTV9.mbn2P1tAOf8-vCEz7tWLEyxvGtDmQ1iS7XPC3YjpW80", "user": { "id": 6, "username": "admin1", "email": "admin1@example.com","role": "admin", "is_active": true }}

POST /logout

Description: Invalidate the user's session/token on the client side.

Authentication: Token required.

Example Success Response (200 OK):

{"message": "Logged out successfully"}

POST /forgot-password

Description: Verifies the username exists in the database to initiate recovery.

Authentication: Public.

Request Body Parameters:

{ "email": "community1@example.com"}

Example Success Response (200 OK):

{"message": "Password reset email generated for community1@example.com", "preview": "https://ethereal.email/message/agCkF77r6kD8o-ruagCkGoIKErNPYURYAAAAAZk0mQw1qcbBmKViHnTYGgI"}

POST /reset-password

Description: Updates the password for a specific user after verification.

Authentication: Public.

Request Body Parameters

{ "email": "community1@example.com", "newPassword": "654321"}

Example Success Response (200 OK):

{ "message": "Password reset successfully", "user": { "user_id": 3, "username": "community1", "email": "community1@example.com", "role": "community_member" }}

Role-Based Access Control (RBAC)

The backend uses a two-gate middleware system to protect routes. This is implemented in authMiddleware.js and applied to routes in issueRoutes.js and userRoutes.js.

Gate 1: verifyToken

Checks the Authorization header for a Bearer <token>.

If missing or invalid, returns 401 Unauthorized.

Gate 2: authorizeRoles (...allowedRoles)

Extracts the role from the decoded JWT.

If the user's role is not in the allowedRoles list, returns 403 Forbidden.

2. Issue Management APIs

POST http://localhost:5000/api/issues

Description: Submit a new issue.

Authorization: Bearer COMMUNITY_TOKEN

Request Body Parameters: { "title": "Broken chair in lecture hall", "description": "One of the chairs is broken and unsafe to use.", "location_id": 1, "category_id": 3}

Example Success Response (201 OK):

{ "message": "Issue created successfully", "issue": { "ticket_id": 1, "title": "Broken chair in lecture hall", "description": "One of the chairs is broken and unsafe to use.", "photo_url": null, "completed_photo_url": null, "location_id": 1, "category_id": 3, "status": "pending", "priority": "medium" "created_by": 3, "assigned_to": null, "created_at": "2026-05-10T12:32:53.282Z","updated_at": "2026-05-10T12:32:53.282Z" }}

GET http://localhost:5000/api/issues

Description: Get all issues

Authorization: Bearer MANAGER_TOKEN

Example Success Response (200 OK):

{ "count": 1, "issues": [{ "ticket_id": 1, "title": "Broken chair in lecture hall", "description": "One of the chairs is broken and unsafe to use.", "photo_url": null, "completed_photo_url": null, "location_id": 1, "category_id": 3, "status": "pending", "priority": "medium", "created_by": 3, "assigned_to": null, "created_at": "2026-05-10T12:32:53.282Z", "updated_at": "2026-05-10T12:32:53.282Z" }]}

GET http://localhost:5000/api/issues/my

Description: Get issues submitted by the logged-in Community Member

Authorization: Bearer COMMUNITY_TOKEN

Example Success Response (200 OK):

{ "count": 1, "issues": [{ "ticket_id": 1, "title": "Broken chair in lecture hall", "description": "One of the chairs is broken and unsafe to use.", "photo_url": null, "completed_photo_url": null, "location_id": 1, "category_id": 3, "status": "pending", "priority": "medium",

"created_by": 3, "assigned_to": null, "created_at": "2026-05-10T12:32:53.282Z", "updated_at": "2026-05-10T12:32:53.282Z" }]}

GET http://localhost:5000/api/issues/assigned/my

Description: get all assigned issues

Authorization: Bearer WORKER_TOKEN

Example Success Response (200 OK):

{ "count": 0, "issues": []}

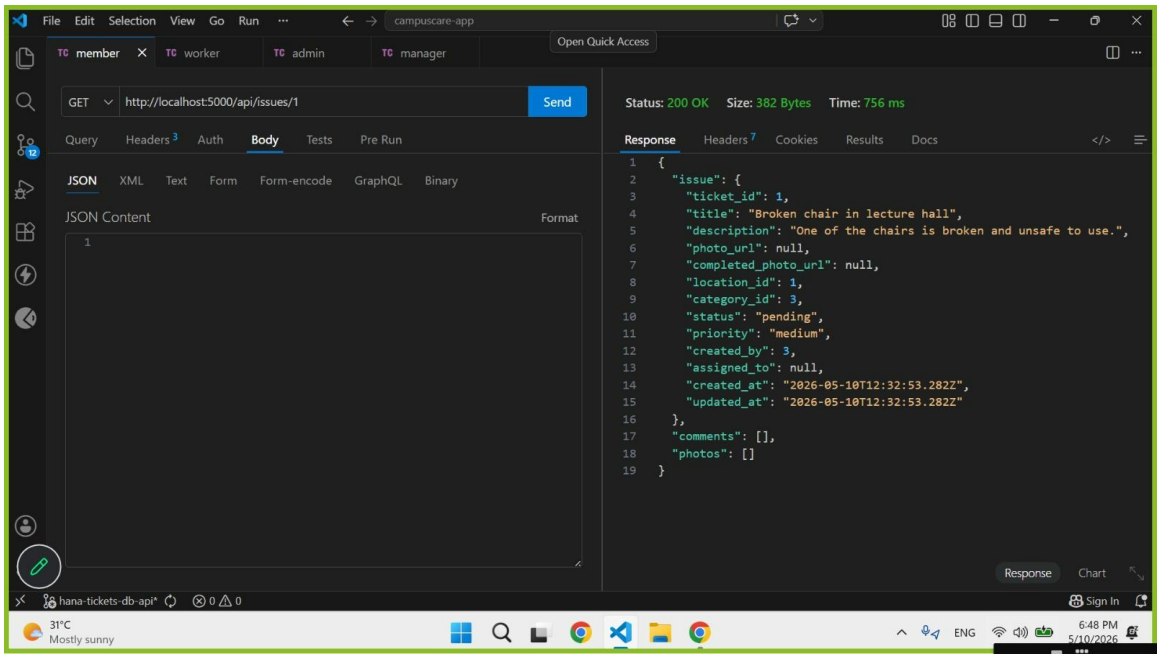
GET http://localhost:5000/api/issues/1

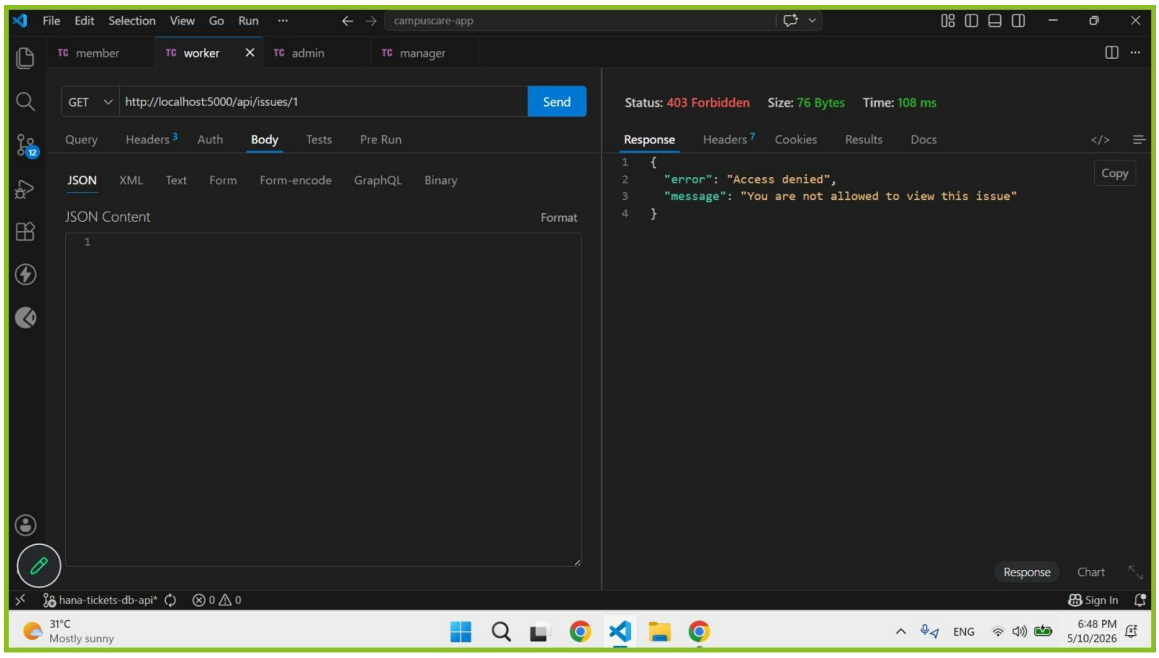
Description: Facility Manager: can view all issues.

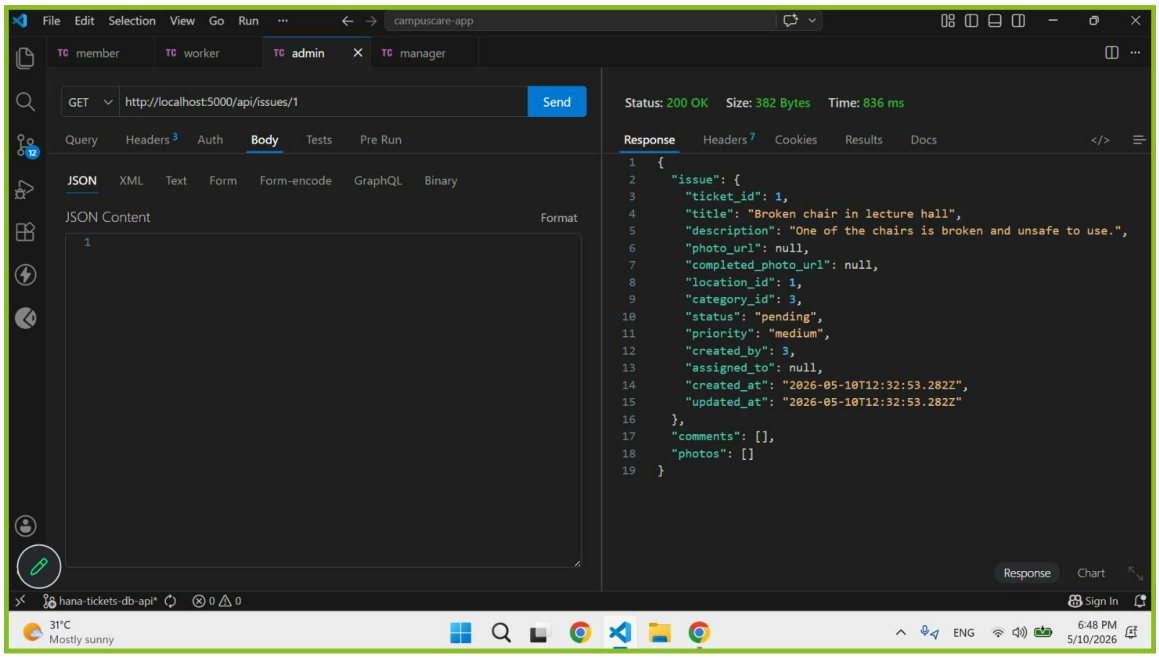
Admin: can view all issues.

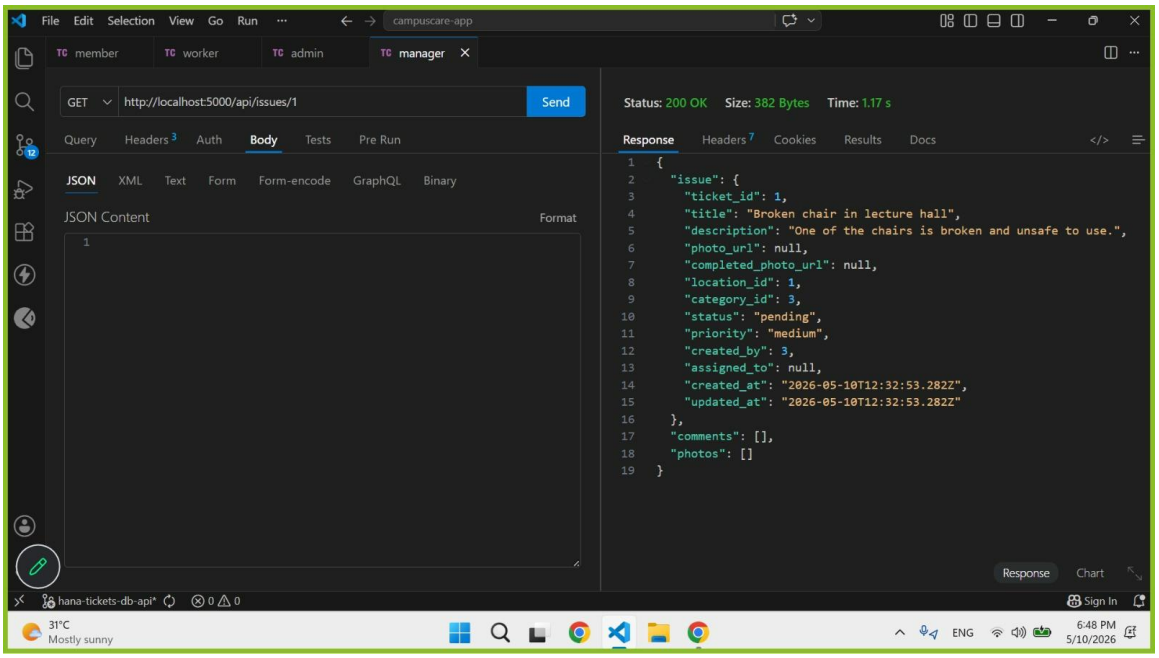
Community Member: can view only their own submitted issues.

Worker: can view only issues assigned to them.




File Edit Selection View Go Run
campuscare-app
member
worker
admin
manager
GET
http://localhost:5000/api/issues/1
Send
Query
Headers 3
Auth
Body
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
Status: 403 Forbidden
Size: 76 Bytes
Time: 108 ms
Response
Headers 7
Cookies
Results
Docs
Copy
{
"error": "Access denied",
"message": "You are not allowed to view this issue"
}
Response
Chart
hana-tickets-db-api*
Sign In
31°C
Mostly sunny
ENG
6:48 PM
5/10/2026


File Edit Selection View Go Run
campuscare-app
member
worker
admin
manager
GET
http://localhost:5000/api/issues/1
Send
Query
Headers 3
Auth
Body
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
Status: 200 OK
Size: 382 Bytes
Time: 836 ms
Response
Headers 7
Cookies
Results
Docs
{
"issue": {
"ticket_id": 1,
"title": "Broken chair in lecture hall",
"description": "One of the chairs is broken and unsafe to use.",
"photo_url": null,
"completed_photo_url": null,
"location_id": 1,
"category_id": 3,
"status": "pending",
"priority": "medium",
"created_by": 3,
"assigned_to": null,
"created_at": "2026-05-10T12:32:53.282Z",
"updated_at": "2026-05-10T12:32:53.282Z"
},
"comments": [],
"photos": []
}
Response
Chart
hana-tickets-db-api*
Sign In
31°C
Mostly sunny
ENG
6:48 PM
5/10/2026



PUT http://localhost:5000/api/issues/1/assign

Description: Assign an issue to a worker (Facility Manager)

Request Body Parameters: { "assigned_to": 5}

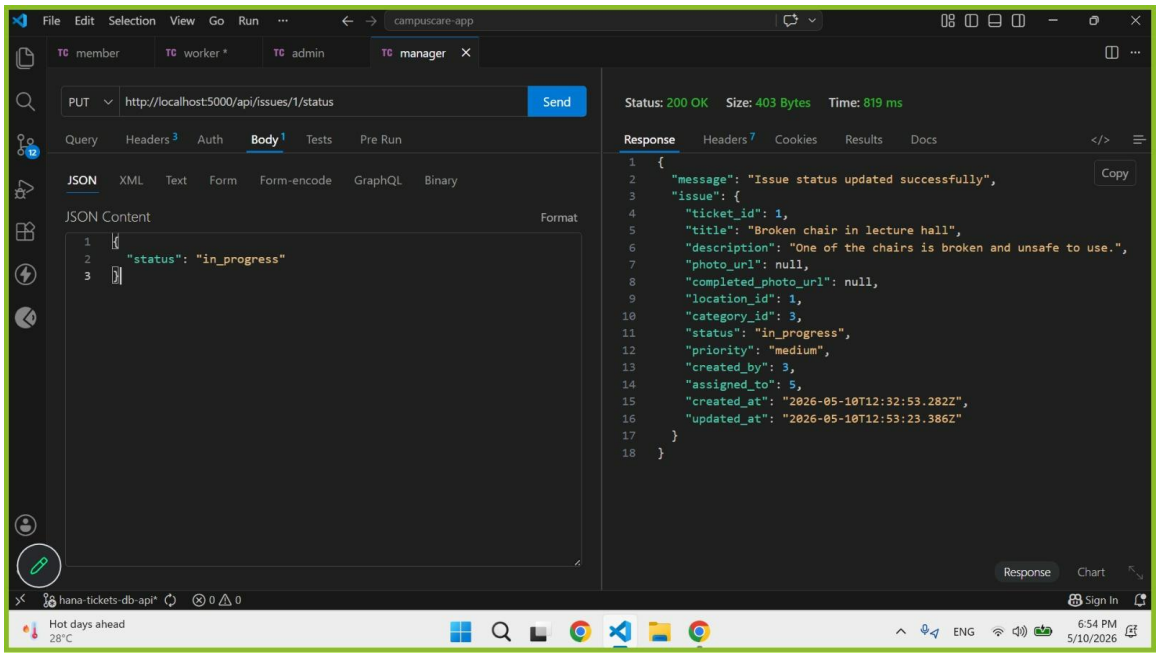
Authorization: Bearer MANAGER_TOKEN

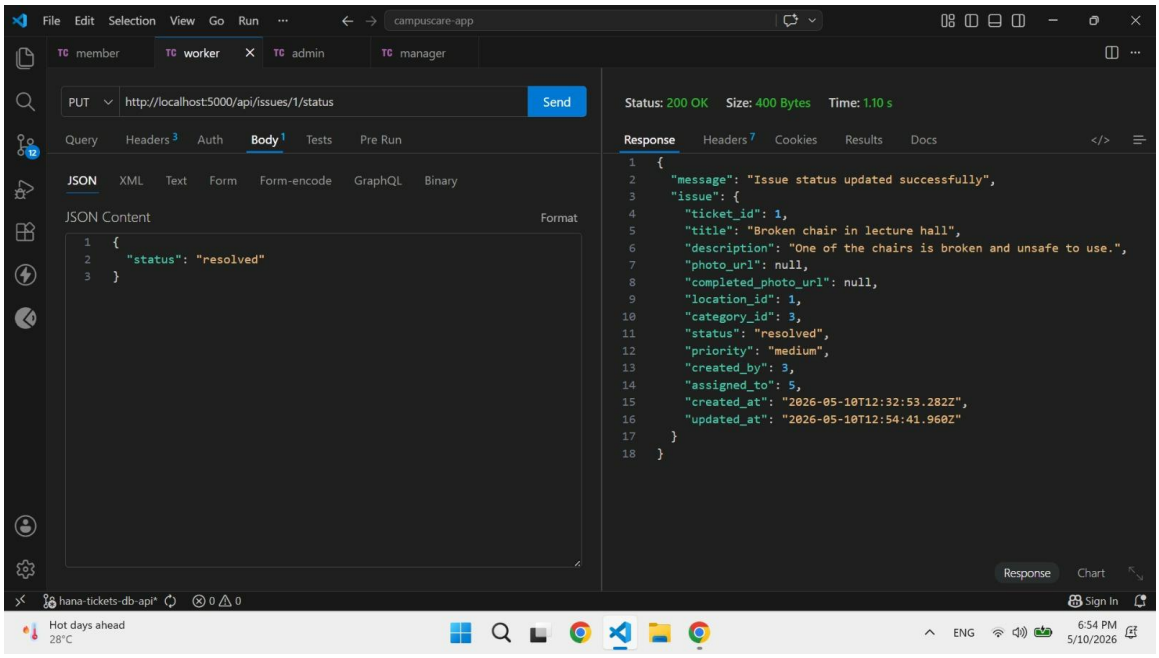
Example Success Response (200 OK):

{"message": "Issue assigned successfully", "issue": { "ticket_id": 1, "title": "Broken chair in lecture hall", "description": "One of the chairs is broken and unsafe to use.", "photo_url": null, "completed_photo_url": null, "location_id": 1, "category_id": 3, "status": "assigned", "priority": "medium", "created_by": 3, "assigned_to": 5, "created_at": "2026-05-10T12:32:53.282Z", "updated_at": "2026-05-10T12:50:57.000Z" }}

PUT http://localhost:5000/api/issues/1/status

Description: Facility Manager can update status ,Worker can only update assigned issues.,Worker can only set status to in_progress or resolved





PUT http://localhost:5000/api/issues/1/priority

Description: update issue priority

Request Body Parameters { "priority": "high"}

Authorization: Bearer MANAGER_TOKEN

Example Success Response (200 OK):

{ "message": "Issue priority updated successfully","issue": { "ticket_id": 1, "title": "Broken chair in lecture hall","description": "One of the chairs is broken and unsafe to use.", "photo_url": null, "completed_photo_url": null, "location_id": 1, "category_id": 3, "status": "resolved", "priority": "high", "created_by": 3, "assigned_to": 5, "created_at": "2026-05-10T12:32:53.282Z", "updated_at": "2026-05-10T12:56:33.337Z"}}

PUT http://localhost:5000/api/issues/1/close

Description: Close a resolved issue (Facility Manager)

Authorization: Bearer MANAGER_TOKEN

Example Success Response (200 OK):

{ "message": "Issue closed successfully", "issue": { "ticket_id": 1, "title": "Broken chair in lecture hall", "description": "One of the chairs is broken and unsafe to use.", "photo_url": null, "completed_photo_url": null,"location_id": 1,"category_id": 3, "status": "closed", "priority": "high","created_by": 3,"assigned_to": 5,"created_at": "2026-05-10T12:32:53.282Z","updated_at": "2026-05-10T12:58:01.843Z" }}

POST http://localhost:5000/api/issues/1/comments

Description: Add a comment to an issue (Worker)

Request Body Parameters: { "comment_text": "I checked the issue and started fixing it."}

Authorization: Bearer WORKER_TOKEN

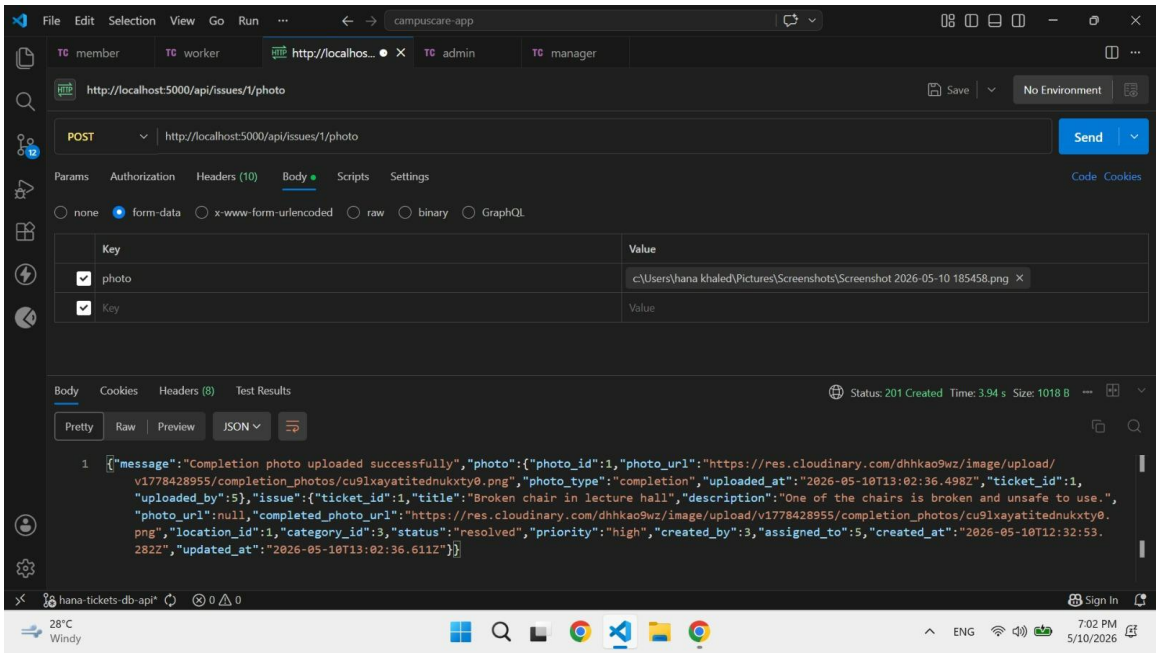
Example Success Response (201 OK):

{ "message": "Comment added successfully", "comment": {"comment_id": 1, "comment_text": "I checked the issue and started fixing it.", "created_at": "2026-05-10T13:00:20.092Z", "ticket_id": 1, "user_id": 5 }}

POST http://localhost:5000/api/issues/1/photo

Description: Upload a completion photo (Worker)

Authorization: Bearer WORKER_TOKEN



DELETE http://localhost:5000/api/issues/1

Description: Delete an issue (Facility Manager)

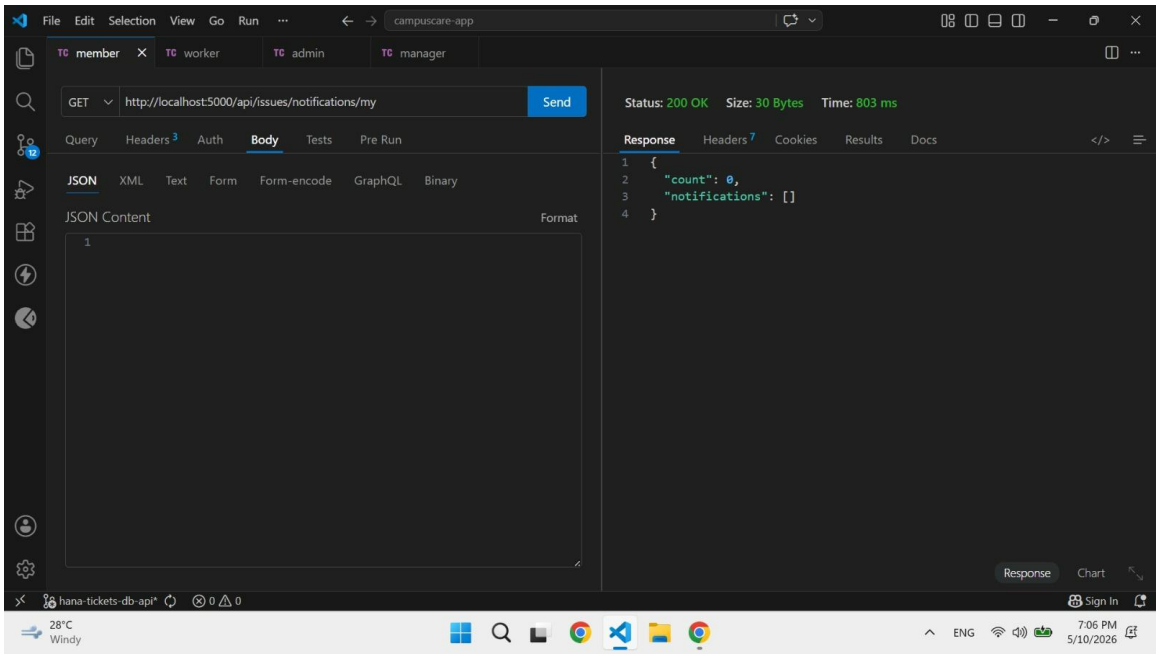
Authorization: Bearer MANAGER_TOKEN

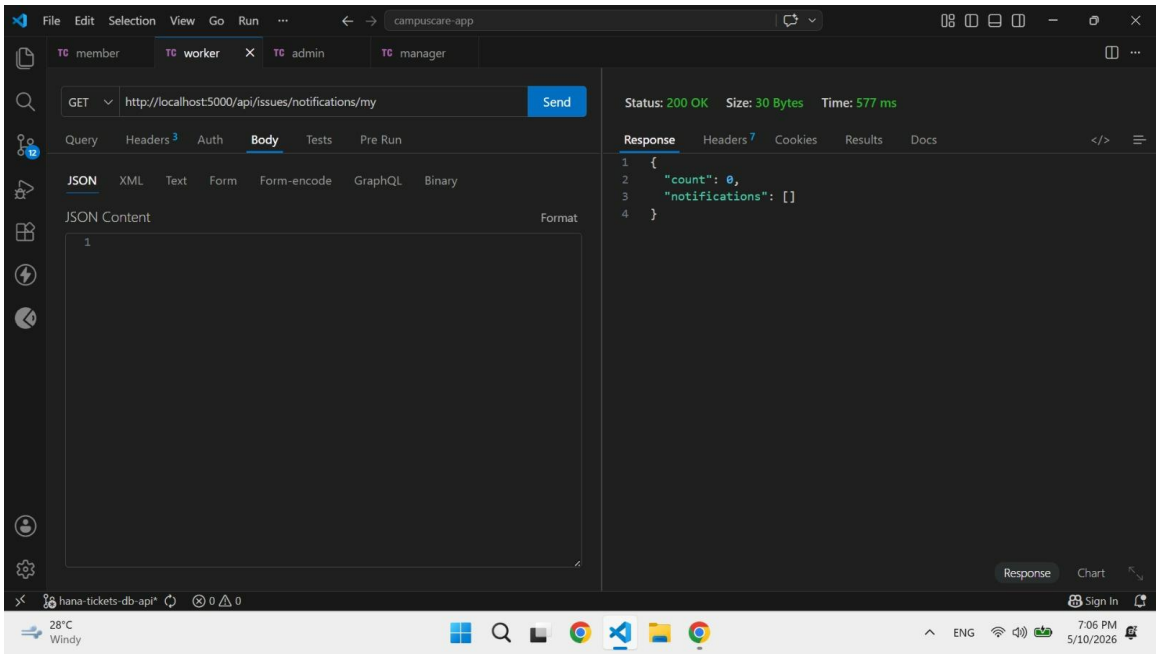
Example Success Response (200 OK):

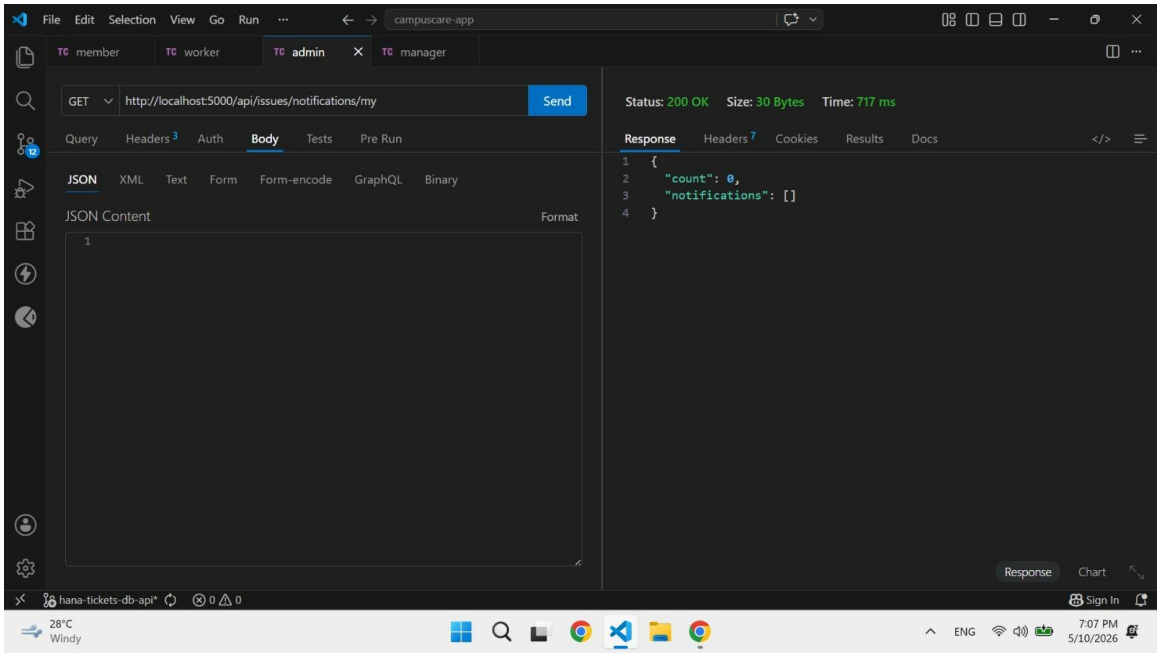
{ "message": "Issue deleted successfully", "deletedIssue": { "ticket_id": 1, "title": "Broken chair in lecture hall","description": "One of the chairs is broken and unsafe to use.", "photo_url": null, "completed_photo_url": "https://res.cloudinary.com/dhhkao9wz/image/upload/v1778428955/completion_photos/cu9lxayatitednukxty0.png", "location_id": 1, "category_id": 3,"status": "resolved", "priority": "high","created_by": 3,"assigned_to": 5, "created_at": "2026-05-10T12:32:53.282Z", "updated_at": "2026-05-10T13:02:36.611Z" }}

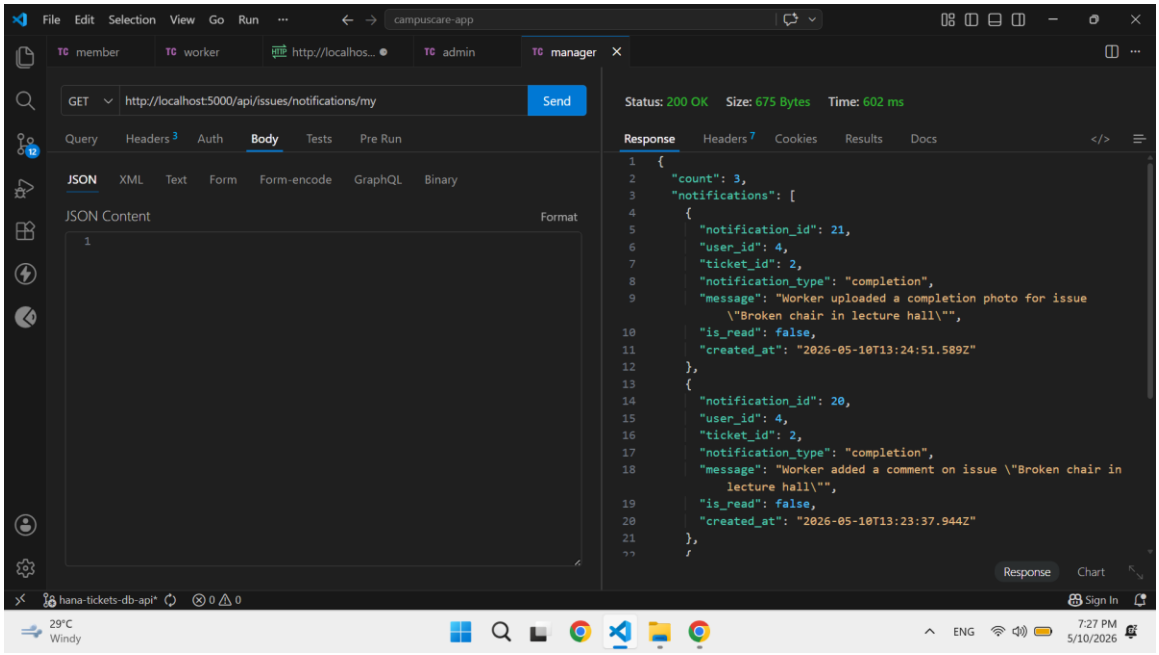
GET http://localhost:5000/api/issues/notifications/my

Description: get any login user notification



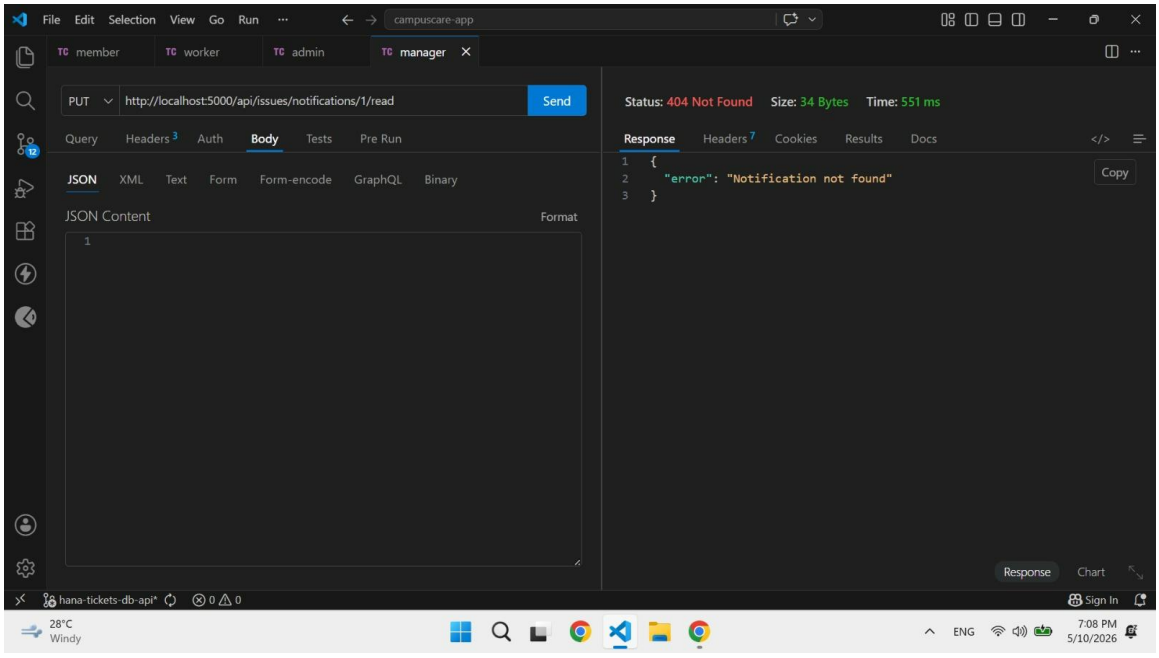


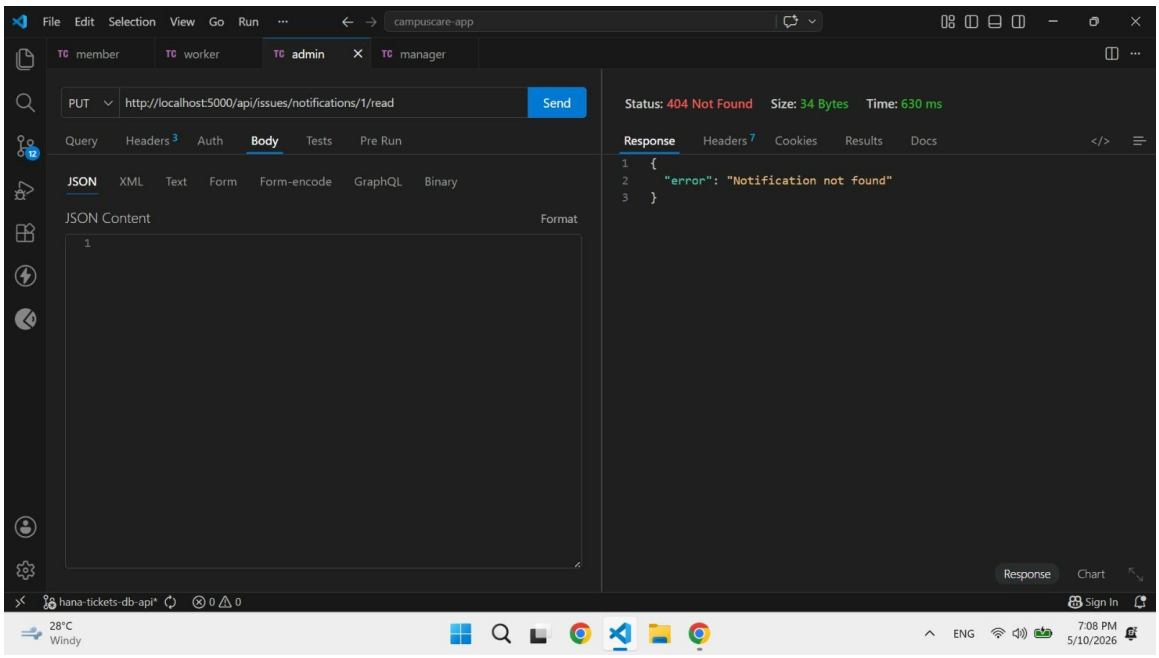

File Edit Selection View Go Run
campuscare-app
member
worker
admin
manager
GET
http://localhost:5000/api/issues/notifications/my
Send
Query
Headers 3
Auth
Body
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
Status: 200 OK
Size: 30 Bytes
Time: 717 ms
Response
Headers 7
Cookies
Results
Docs
{
"count": 0,
"notifications": []
}
Response
Chart
hana-tickets-db-api*
Sign In
28°C
Windy
ENG
7:07 PM
5/10/2026

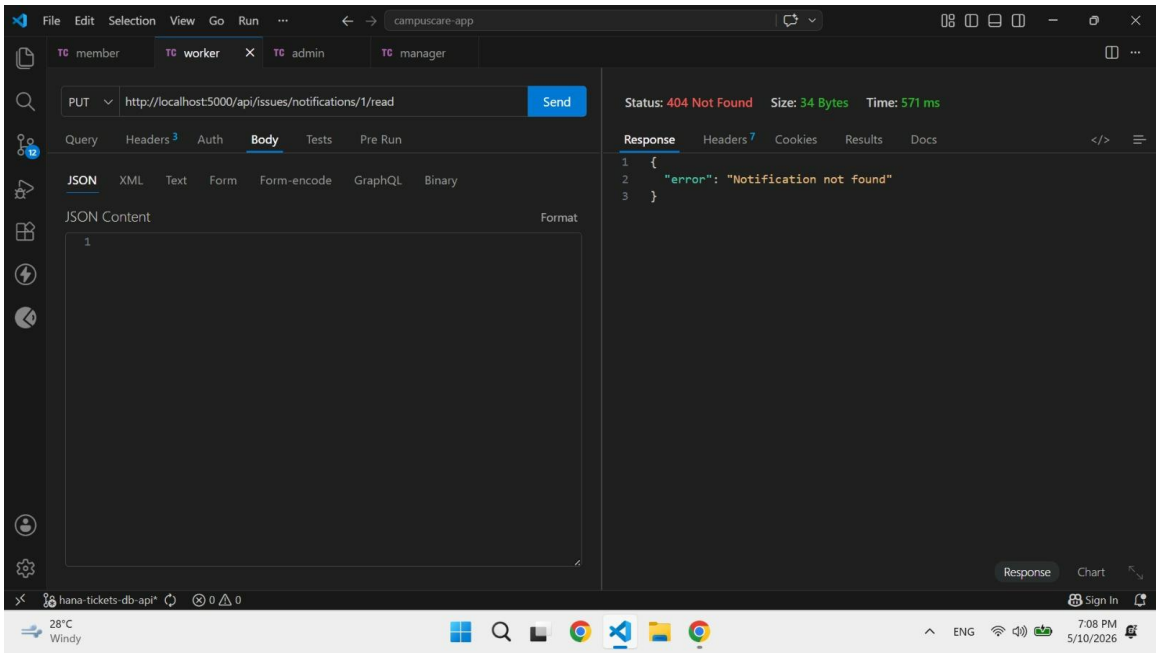

File Edit Selection View Go Run
campuscare-app
member
worker
http://localhos...
admin
manager
GET
http://localhost:5000/api/issues/notifications/my
Send
Query
Headers 3
Auth
Body
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
Status: 200 OK
Size: 675 Bytes
Time: 602 ms
Response
Headers 7
Cookies
Results
Docs
{
"count": 3,
"notifications": [
{
"notification_id": 21,
"user_id": 4,
"ticket_id": 2,
"notification_type": "completion",
"message": "Worker uploaded a completion photo for issue \"Broken chair in lecture hall\"",
"is_read": false,
"created_at": "2026-05-10T13:24:51.589Z"
},
{
"notification_id": 20,
"user_id": 4,
"ticket_id": 2,
"notification_type": "completion",
"message": "Worker added a comment on issue \"Broken chair in lecture hall\"",
"is_read": false,
"created_at": "2026-05-10T13:23:37.944Z"
},
Response
Chart
hana-tickets-db-api*
Sign In
29°C
Windy
ENG
7:27 PM
5/10/2026

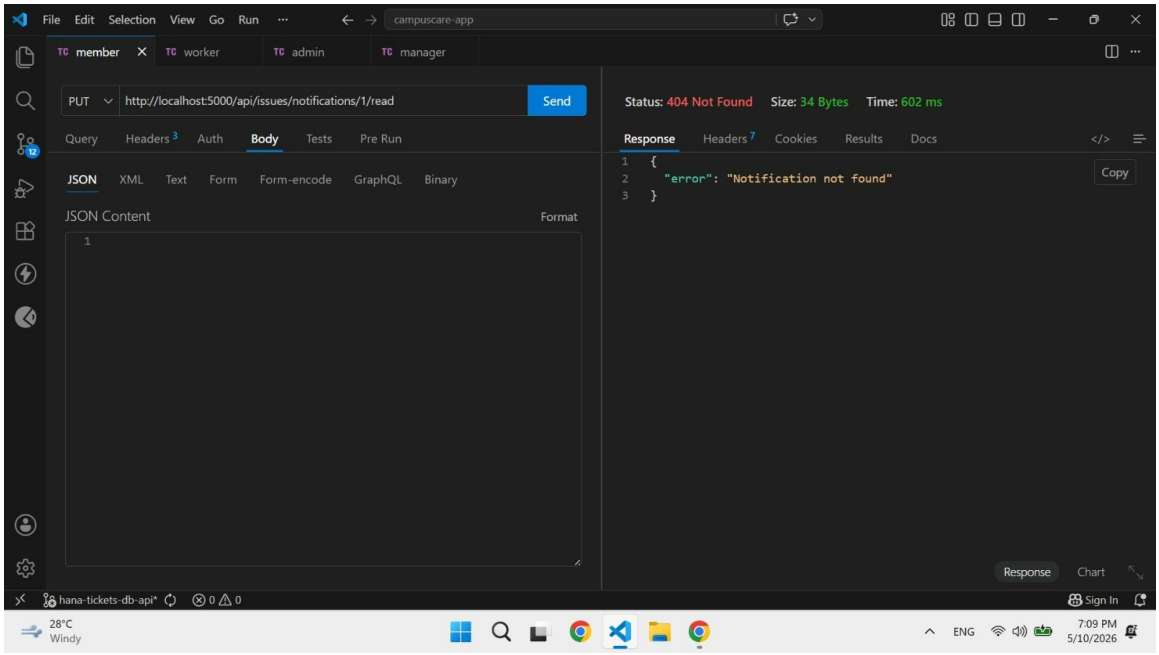
PUT http://localhost:5000/api/issues/notifications/1/read

Description: error since there is no notification right now , the user marks their own notification as read









GET http://localhost:5000/api/manager/workers

Description: List all worker accounts (Facility Manager)

Authorization: Bearer MANAGER_TOKEN

Example Success Response (200 OK):

{ "count": 1, "workers": [{ "user_id": 5,"username": "worker1","email": "worker1@example.com","role": "worker","is_active": true,created_at": "2026-05-10T12:19:14.900Z" }]}

PUT http://localhost:5000/api/manager/workers/5/status

Description: Activate or deactivate a worker account

(Facility Manager)

Authorization: Bearer MANAGER_TOKEN

Request Body Parameters:{ "is_active": false}

Example Success Response (200 OK):

{ "message": "Worker account deactivated successfully", "worker": { "user_id": 5, "username": "worker1", "email": "worker1@example.com","role": "worker", "is_active": false}}

GET http://localhost:5000/api/admin/users

Description: List all registered users

Authorization: Bearer ADMIN_TOKEN

Example Success Response (200 OK):

{ "count": 5,"users": [{"user_id": 6,"username": "admin1","email": "admin1@example.com","role": "admin", "is_active": true, "created_at": "2026-05-10T12:19:47.873Z" },{"user_id": 5, "username": "worker1","email": "worker1@example.com","role": "worker", "is_active": false, "created_at": "2026-05-10T12:19:14.900Z"},{ "user_id": 4, "username": "manager1","email": "manager1@example.com", "role": "facility_manager", "is_active": true,"created_at": "2026-05-10T12:18:35.112Z" }, {"user_id": 3,"username": "community1","email": "community1@example.com", "role": "community_member","is_active": true,"created_at": "2026-05-10T12:17:05.849Z" },{ "user_id": 2,"username": "joy_tester","email": "joy@campuscare.edu", "role": "facility_manager" "is_active": true, "created_at": "2026-05-10T11:51:37.193Z" }]}

PUT http://localhost:5000/api/admin/users/2/status

Description: Activate or deactivate a user account

Authorization: Bearer ADMIN_TOKEN

Request Body Parameters: { "is_active": false}

Example Success Response (200 OK):

{ "message": "User account deactivated successfully","user": {"user_id": 2,"username": "joy_tester", "email": "joy@campuscare.edu","role": "facility_manager","is_active": false}}